



Bill of Materials and Parts List

Celestial Electric Solar & Battery Room Monitor

This parts list supports the Arduino Project Hub build. Store links are examples for the project documentation and do not require buying from a specific supplier.

Recommended sensor choice: Use BME280 as the primary temperature/humidity sensor. DHT20 is an alternate. You do not need both for the finished demo.

Devices and components

Item	Qty	Role	Store link
Arduino UNO R4 WiFi	1	Main Wi-Fi capable microcontroller board for sensor reading and cloud dashboard updates.	Open link
Adafruit BME280 Temperature, Humidity, Pressure Sensor - STEMMA QT	1	Recommended room temperature and humidity sensor. Pressure is optional.	Open link
DHT20 / AHT20 I2C Temperature and Humidity Sensor	1 optional	Alternate temperature/humidity sensor if BME280 is not used.	Open link
STEMMA QT / Qwiic JST SH 4-pin Cable - 100 mm	1	Clean I2C connection between board and STEMMA/Qwiic sensor.	Open link
Magnetic Contact Switch Door Sensor	1	Door or enclosure contact status input.	Open link
Simple Water Detection Sensor with Digital Output	1	Water leak / wet-floor detection input for the demo.	Open link
Super Bright Red 5 mm LED	1	Local alert indication.	Open link
Super Bright Green 5 mm LED	1	Local normal-status indication.	Open link
Through-Hole Resistors - 220 ohm, 5%, 1/4 W	2+	LED current limiting resistors.	Open link
Full Sized Premium Breadboard - 830 Tie Points	1	Prototype wiring and testing.	Open link
Breadboarding Wire Bundle	1	Prototype jumpers and low-voltage leads.	Open link
USB Type A to USB Type C Cable - approx. 1 meter	1	Programming and USB power for the demo.	Open link
Official Raspberry Pi 5.1 V / 3 A USB-C Power Supply	1 optional	Bench power supply for standalone demo use.	Open link
Small Plastic Project Enclosure - Weatherproof with Clear Top	1 optional	Demo enclosure for a cleaner equipment-room monitor look.	Open link

Hardware tools

Tool	Purpose	Link
Multi-size wire stripper and cutter	Cutting and stripping low-voltage conductors.	Open link
38 Piece Screwdriver Set	Small terminal screws and enclosure work.	Open link
Flush diagonal cutters	Trimming component leads and jumpers.	Open link
Digital multimeter	Verifying power, continuity, sensor states, and polarity.	Open link

Software and platforms

Software / platform	Purpose	Link
Arduino IDE	Optional local programming environment.	Open link
Arduino Cloud	Cloud dashboard and IoT variable monitoring.	Open link
Arduino Cloud Editor	Browser-based editor for Arduino sketches.	Open link
Arduino Project Hub	Public project documentation and showcase page.	Open link
GitHub	Code repository and project version history.	Open link

Note: The Arduino Project Hub component section may show placeholder images for custom items. That is okay as long as the component name and store link are saved.